UNIT 7 — SOUND

Revision notes based on the supplied textbook pages, with examples and key facts for competitive examination practice.

1. WHAT IS SOUND?

• Sound is a form of energy that produces the sensation of hearing.

• Sound is produced by vibrating objects. The vibrating source sets the surrounding medium into vibration, and the disturbance travels as a sound wave.

• Sound needs a material medium such as a solid, liquid or gas. It cannot travel through vacuum because there are no particles to transmit the vibration.

• The Moon has no substantial atmosphere. Therefore, sound from an explosion on the Moon cannot travel through space to an observer on Earth.

2. SPEED OF SOUND IN DIFFERENT MEDIA

• Sound travels by transfer of energy from one particle of the medium to neighbouring particles.

• Sound generally travels fastest in solids, slower in liquids and slowest in gases.

$V_{solid} > V_{liquid} > V_{gas}$

The actual speed also depends on the material and its temperature.

3. NOISE AND ITS EFFECTS

• Noise is unwanted, unpleasant or disturbing sound. Loudness level is commonly expressed in decibels (dB).

• Prolonged exposure to loud noise can damage hearing and may lead to temporary or permanent hearing loss.

• Noise can disturb sleep, reduce concentration and work efficiency, and contribute to irritation, stress and tension.

4. REFLECTION OF SOUND

• When sound waves strike a surface and return into the same medium, the phenomenon is called reflection of sound.

• A reflected sound can be heard clearly when the reflecting surface is sufficiently large and hard.

Applications of reflection

Device / application	How it works
Megaphone / speaking tube	Repeated reflections inside the horn help direct sound towards listeners instead of allowing it to spread widely.
Ear trumpet	The wide opening collects sound; reflections guide it towards the narrower end and the ear.
Stethoscope	The chest piece picks up body sounds; sound travels through the instrument's tubing to the doctor's ears.
Sound boards in halls	Curved reflecting surfaces direct sound towards the audience and help distribute it through the hall.

5. ECHO

• An echo is a distinct repetition of a sound heard after reflection from a distant, hard obstacle.

• The ear generally needs a time gap of about 0.1 second to distinguish the reflected sound from the original sound.

Textbook figure: Your supplied page states that 11 metres is the minimum distance for hearing an echo. Learn this figure if your class is following that page.

Physics check: Using 340 m/s for sound and a 0.1 s time gap, distance = (340 × 0.1) ÷ 2 = 17 m. The difference comes from the assumptions used; follow the value expected by your teacher or exam source.

6. RANGE OF HEARING

• The usual audible frequency range for a young human with normal hearing is approximately 20 Hz to 20,000 Hz (20 kHz).

Category	Frequency range	Meaning / examples
Infrasonic	Below 20 Hz	Below the usual lower limit of human hearing; elephants and some whales use very low-frequency sounds.
Audible	About 20 Hz–20 kHz	Usually audible to humans; range varies with age and individual.
Ultrasonic	Above 20 kHz	Above the usual upper limit of human hearing; dogs and bats hear higher frequencies than humans.

• Children may hear some higher frequencies than adults. The ability to hear high-frequency sounds generally decreases with age.

• The human ear is not equally sensitive to every audible frequency; it is particularly sensitive in the region of a few thousand hertz.

7. EXAMPLES OF ANIMAL HEARING

Animal	Approximate hearing fact from the supplied text
Dog	Can hear frequencies up to about 50 kHz.
Bat	Can hear frequencies up to about 100 kHz.
Dolphins	Can hear sounds at frequencies higher than the human audible range.
Elephants and some whales	Can detect very low-frequency sounds, including infrasound.

These values are approximate and vary by species; treat them as textbook comparisons, not exact limits for every individual.

8. PRACTICAL: DROP A STONE FROM DIFFERENT HEIGHTS

• Drop the same stone onto the same table from two different heights, such as 20 cm and 100 cm.

• The stone dropped from the greater height generally strikes the table with greater speed and kinetic energy, producing a louder impact sound.

• Keep the stone, table and surroundings the same for a fair comparison. Change only the drop height.

9. QUICK REVISION FACTS

• Sound is produced by vibrations; sound needs a material medium and cannot travel through vacuum.

• Speed order: solid > liquid > gas.

• Noise level is measured in decibels (dB).

• Reflection means sound returns into the same medium after striking a surface.

• Human audible range: approximately 20 Hz–20 kHz.

• Infrasound: below 20 Hz; ultrasound: above 20 kHz.

• Dog: approximately 50 kHz; bat: approximately 100 kHz, as approximate textbook values.

Exam tip: Distinguish frequency (Hz) from loudness level (dB), and echo (reflected sound) from noise (unwanted sound).